

# Assignment 03 (FEM)

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## 1 Question 1

Poisson system:

$$-\nabla^2 u = f \quad (1)$$

$$f(x, y) = 2\pi^2 \sin(\pi x) \cos(\pi y) \quad (2)$$

$$u(0, y) = u(1, y) = 0 \quad (3)$$

$$\partial_y u|_{y=0} = \partial_y u|_{y=1} = 0 \quad (4)$$

exact solution is

$$u_e(x, y) = \sin(\pi x) \cos(\pi y) \quad (5)$$

To verify this exact solution,

$$\nabla^2 u_e = u_{e,xx} + u_{e,yy} = -f(x, y) \quad (6)$$

Therefore, it becomes,

$$-\nabla^2 u_e = \pi^2 \sin(\pi x) \cos(\pi y) + \pi^2 \sin(\pi x) \cos(\pi y) = 2\pi^2 \sin(\pi x) \cos(\pi y) = f \quad (7)$$

Boundary conditions:

1. Dirichlet Boundary Condition

$$u_e(0, y) = \sin(0) \cos(\pi y) = 0 \quad (8)$$

$$u_e(1, y) = \sin(\pi) \cos(\pi y) = 0 \quad (9)$$

2. Neumann Boundary Condition

$$\partial_y u_e|_{y=0} = -\pi \sin(\pi x) \sin(0) = 0 \quad (10)$$

$$\partial_y u_e|_{y=1} = -\pi \sin(\pi x) \sin(\pi) = 0 \quad (11)$$

justifies that this is exact solution of poisson equation

## 1.1

### 1. Ritz-Galerkin principle

Solution of Poisson Equation is equal to the minimisation of the energy function

$$J(u) = \frac{1}{2} \int_{\Omega} |\nabla u|^2 d\Omega - \int_{\Omega} f u d\Omega$$

and here  $\Omega = [0, 1]^2$

As Dirichlet boundary conditions are imposed at  $x=0$  and  $x=1$ , therefore the functional space should be,

$$V = u : u(0, y) = u(1, y) = 0$$

Consider a perturbed function

$$u_{\varepsilon} = u + \varepsilon \delta u$$

where  $\varepsilon$  is a small scalar parameter. Substitute this into the Energy

$$J(u + \varepsilon \delta u) = \frac{1}{2} \int_{\Omega} |\nabla(u + \varepsilon \delta u)|^2 d\Omega - \int_{\Omega} f(u + \varepsilon \delta u) d\Omega$$

Since

$$\nabla(u + \varepsilon \delta u) = \nabla u + \varepsilon \nabla(\delta u)$$

we obtain

$$|\nabla u + \varepsilon \nabla \delta u|^2 = |\nabla u|^2 + 2\varepsilon \nabla u \cdot \nabla \delta u + \varepsilon^2 |\nabla \delta u|^2$$

Expanded functional form is;

$$J(u + \varepsilon \delta u) = \frac{1}{2} \int_{\Omega} (|\nabla u|^2 + 2\varepsilon \nabla u \cdot \nabla \delta u + \varepsilon^2 |\nabla \delta u|^2) d\Omega - \int_{\Omega} f u d\Omega - \varepsilon \int_{\Omega} f \delta u d\Omega$$

Differentiating with respect to  $\varepsilon$  for  $\lim_{\varepsilon \rightarrow 0}$

$$\delta J(u) = \left. \frac{d}{d\varepsilon} J(u + \varepsilon \delta u) \right|_{\varepsilon=0} = \int_{\Omega} \nabla u \cdot \nabla \delta u d\Omega - \int_{\Omega} f \delta u d\Omega$$

and

$$\delta J(u) = 0$$

Therefore,

$$\int_{\Omega} \nabla u \cdot \nabla \delta u \, d\Omega = \int_{\Omega} f \delta u \, d\Omega$$

Integration by Parts

$$\int_{\Omega} \nabla u \cdot \nabla \delta u \, d\Omega = - \int_{\Omega} (\nabla^2 u) \delta u \, d\Omega + \int_{\partial\Omega} \delta u \frac{\partial u}{\partial n} \, ds \, d\Omega$$

we get

$$- \int_{\Omega} (\nabla^2 u) \delta u \, d\Omega + \int_{\partial\Omega} \delta u \frac{\partial u}{\partial n} \, d\Omega = \int_{\Omega} f \delta u \, d\Omega$$

After rearrangement,

$$\int_{\Omega} (-\nabla^2 u - f) \delta u \, d\Omega + \int_{\partial\Omega} \delta u \frac{\partial u}{\partial n} \, d\Omega = 0$$

applying the Boundary Conditions,  
for Dirichlet boundary,  $u$  is fixed, therefore

$$\delta u(0, y) = \delta u(1, y) = 0$$

And for Neumann boundary,

$$\frac{\partial u}{\partial n} = 0$$

Thus we obtain

$$\int_{\Omega} (-\nabla^2 u - f) \delta u \, d\Omega = 0$$

Since  $\delta u$  is arbitrary, we recover the strong form

$$-\nabla^2 u = f$$

## 2. Weak Formulation

Returning to the

$$\int_{\Omega} \nabla u \cdot \nabla \delta u \, d\Omega = \int_{\Omega} f \delta u \, d\Omega$$

and substituting

$$w = \delta u$$

here  $w(x, y)$  is the test function. And the obtained weak formulation is;

$$\boxed{\int_{\Omega} \nabla u \cdot \nabla w \, d\Omega = \int_{\Omega} f w \, d\Omega \quad \forall w \in V}$$

$V$  is the test space = where  $\{w(x, y) = 0 \text{ on Dirichlet boundary}\}$

This is the same form as obtained from the Ritz-Galerkin method.

## 1.2 Algebraic or discrete Ritz-Galerkin principle

Choosing a finite-dimensional subspace  $V_h$  from an infinite-dimensional space  $V$ , which is spanned by basis functions:

$$V_h = \text{span}\{\phi_1, \phi_2, \dots, \phi_N\}$$

Each basis function  $\phi_i(x, y)$  satisfies the Dirichlet boundary condition. And  $u_j$  are unknown coefficients.

Therefore for the approximate solution,

$$u_h(x, y) = \sum_{j=1}^N u_j \phi_j(x, y)$$

The global basis function is  $w = \phi_i$ , choosing the test function in the same space.

Now, substituting into the weak form, that gives,

$$\int_{\Omega} \nabla \left( \sum_{j=1}^N u_j \phi_j \right) \cdot \nabla \phi_i \, d\Omega = \int_{\Omega} f \phi_i \, d\Omega$$

and this can also be written as,

$$\sum_{j=1}^N u_j \int_{\Omega} \nabla \phi_j \cdot \nabla \phi_i \, d\Omega = \int_{\Omega} f \phi_i \, d\Omega$$

The matrix representation for this,

$$K_{ij} = \int_{\Omega} \nabla \phi_j \cdot \nabla \phi_i \, d\Omega$$

and

$$F_i = \int_{\Omega} f \phi_i \, d\Omega$$

Therefore, the discrete system becomes

$$\sum_{j=1}^N K_{ij} u_j = F_i$$

And the matrix form

$$KU = F$$

### Discrete Ritz Principle

$$J(u) = \frac{1}{2} \int_{\Omega} |\nabla u|^2 d\Omega - \int_{\Omega} f u d\Omega$$

we are approximating with,

$$u_h(x, y) = \sum_{j=1}^N U_j N_j(x, y)$$

After substitution;

$$J(u_h) = \frac{1}{2} \int_{\Omega} \left| \nabla \sum_{j=1}^N U_j N_j \right|^2 d\Omega - \int_{\Omega} f \sum_{j=1}^N U_j N_j d\Omega$$

it can be written as,

$$J(u_h) = \frac{1}{2} \int_{\Omega} \left( \sum_{i=1}^N U_i \nabla N_i \right) \cdot \left( \sum_{j=1}^N U_j \nabla N_j \right) d\Omega - \sum_{j=1}^N U_j \int_{\Omega} f N_j d\Omega$$

and

$$J(u_h) = \frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N U_i U_j \int_{\Omega} \nabla N_i \cdot \nabla N_j d\Omega - \sum_{j=1}^N U_j \int_{\Omega} f N_j d\Omega$$

For matrix form we define

$$K_{ij} = \int_{\Omega} \nabla N_i \cdot \nabla N_j \text{ and } F_j = \int_{\Omega} f N_j$$

The discrete form becomes

$$J_h = \frac{1}{2} \sum_{i,j=1}^N U_i U_j K_{ij} - \sum_{j=1}^N U_j F_j$$

According to the Ritz principle the approximate solution is minimized,

$$\frac{\partial J_h}{\partial U_i} = 0$$

Therefore, it becomes

$$\sum_{j=1}^N K_{ij} U_j - F_i = 0$$

Which is same as the previous one, the discrete weak form

$$KU = F$$

### 1.3 Local coordinate system and reference coordinate

For the quadrilateral elements,

$$\Omega = \bigcup_{e=1}^{N_e} \Omega_e$$

$$K_{IJ} = \sum_{e=1}^{N_e} K_{ij}^{(e)}$$

$(I, J)$  are global node indices and  $(i, j)$  local element indices

Quadrilateral elements having coordinates  $(\xi, \eta)$  (local coordinates) are mapped from the standard reference square  $\hat{\Omega} = [-1, 1] \times [-1, 1]$ .

Therefore, the shape functions on the reference element are

$$N_1(\xi, \eta) = \frac{1}{4}(1 - \xi)(1 - \eta)$$

$$N_2(\xi, \eta) = \frac{1}{4}(1 + \xi)(1 - \eta)$$

$$N_3(\xi, \eta) = \frac{1}{4}(1 + \xi)(1 + \eta)$$

$$N_4(\xi, \eta) = \frac{1}{4}(1 - \xi)(1 + \eta)$$

Each element is 1 at its own corner and 0 at other three corners.

Transforming the physical coordinates  $(x, y)$  to the reference coordinates using the shape functions,

$$x(\xi, \eta) = \sum_{i=1}^4 x_i N_i(\xi, \eta)$$

$$y(\xi, \eta) = \sum_{i=1}^4 y_i N_i(\xi, \eta)$$

The Jacobian of this is

$$J = \begin{bmatrix} \frac{\partial x}{\partial \xi} & \frac{\partial x}{\partial \eta} \\ \frac{\partial y}{\partial \xi} & \frac{\partial y}{\partial \eta} \end{bmatrix}$$

Therefore,

$$d\Omega = |J| d\xi d\eta$$

Shape-function gradients in physical space are transformed to

$$\nabla N_i = J^{-T} \nabla_{\xi} N_i$$

where

$$\nabla_{\xi} N_i = \begin{bmatrix} \frac{\partial N_i}{\partial \xi} \\ \frac{\partial N_i}{\partial \eta} \end{bmatrix}$$

Thus

$$\nabla N_i \cdot \nabla N_j = \nabla_{\xi} N_i^T (J^{-1} J^{-T}) \nabla_{\xi} N_j$$

Assuming the metric tensor as

$$G = J^{-1} J^{-T}$$

The local element stiffness matrix is:

$$K_{ij}^{(e)} = \int_{\hat{\Omega}} \nabla N_i \cdot \nabla N_j d\Omega$$

Therefore, the stiffness matrix for element  $e$  becomes

$$K_{ij}^{(e)} = \int_{\hat{\Omega}} (\nabla_{\xi} N_i^T) G (\nabla_{\xi} N_j) (|J| d\xi d\eta)$$

Similarly, the local load vector is,

$$F_i^{(e)} = \int_{\hat{\Omega}} f N_i d\Omega$$

Therefore, it becomes,

$$F_i^{(e)} = \int_{\hat{\Omega}} f(\xi, \eta) N_i(\xi, \eta) (|J| d\xi d\eta)$$

Removing the integration and converting it into a summation with  $n_q$  points,

$$K_{ij}^{(e)} \approx \sum_{q=1}^{n_q} w_q [\nabla_{\xi} N_i^T G \nabla_{\xi} N_j]_q |J|_q$$

and

$$F_i^{(e)} \approx \sum_{q=1}^{n_q} w_q f_q N_i(\xi_q, \eta_q) |J|_q$$

So, for the global system is built by adding all local element contributions using the connectivity mapping:

$$K_{IJ} = \sum_{e=1}^{N_e} K_{ij}^{(e)}, \quad F_I = \sum_{e=1}^{N_e} F_i^{(e)}$$

And the global system is

$$\mathbf{KU} = \mathbf{F}$$

## 1.4 Firedrake solution

Figure 1, 2 and 3 presents the contour plots for  $p = 1, 2$  and  $3$  respectively. Here  $p$  is the polynomial degree of the finite element basis functions. Figure 4 presents the L2error (how far is the numerical solution from the exact solution) vs mesh resolutions for both the methods for  $p = 1, 2$  and  $3$ .

## 1.5 Commented code

Figure 5 shows the section of poisson's equation code with comments.

## 1.6 Modified boundary conditions

We are changing the boundary conditions to understand this better. The new exact solution is changed to

$$u(x, y) = e^x \cos(\pi y)$$



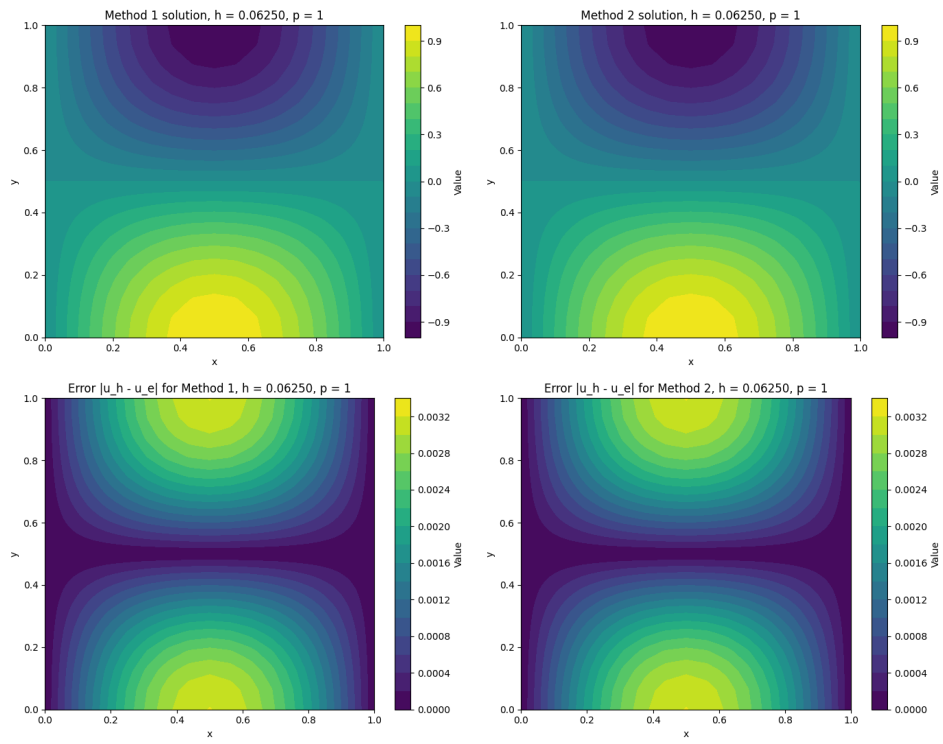


Figure 1: for  $p=1$

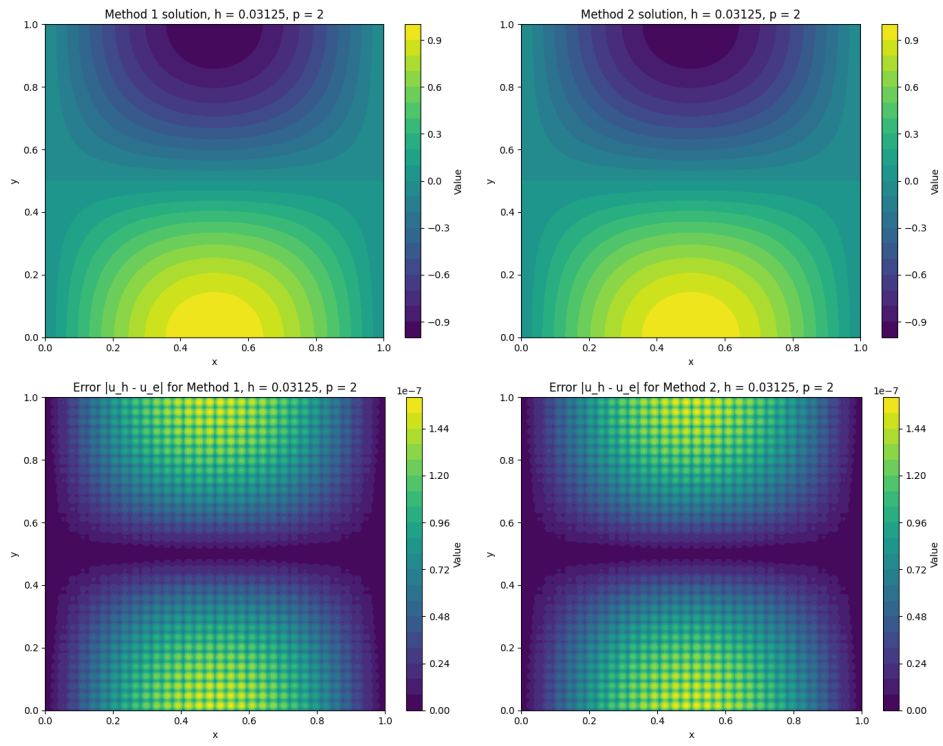


Figure 2: for  $p=2$

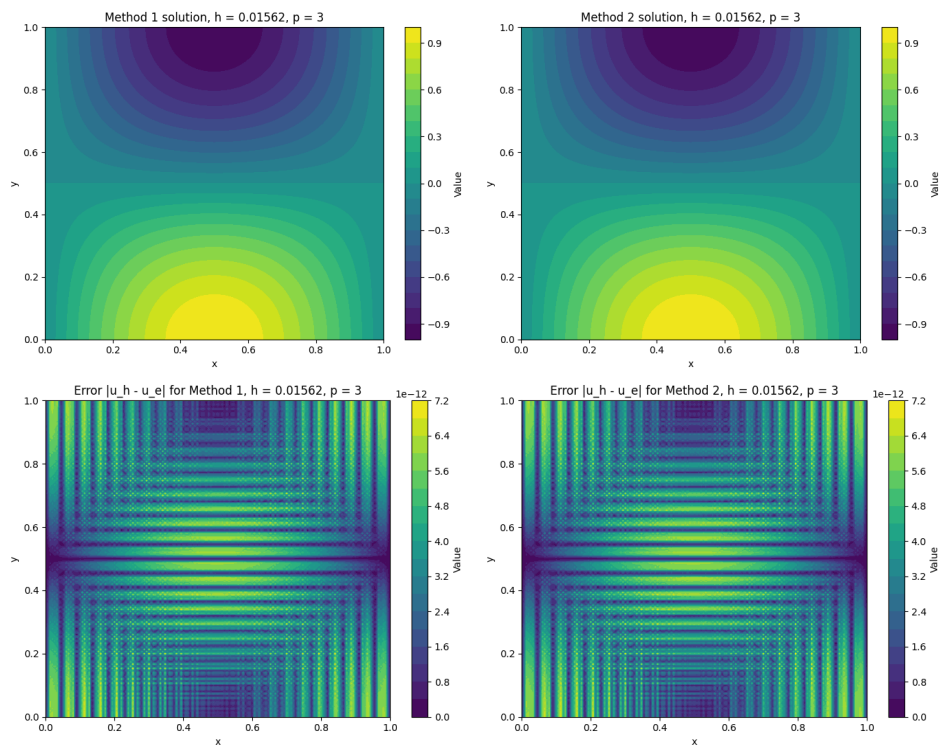


Figure 3: for  $p=3$

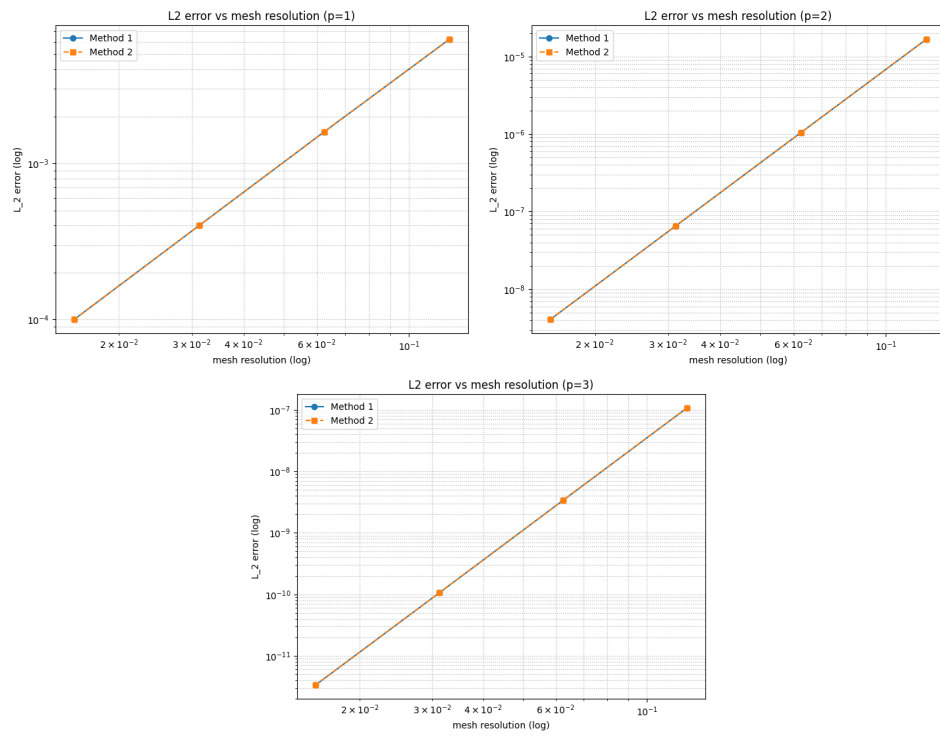


Figure 4: Error relation with the mesh resolutions for both the methods.

```

def solve_poisson_case(nx, p):
    ny = nx
    h = 1.0 / nx

    mesh = UnitSquareMesh(nx, ny, quadrilateral=True)

    # Continuous Galerkin function space of degree p
    V = FunctionSpace(mesh, "CG", p)

    # symbolic coordinates or the mesh coordinates
    x, y = SpatialCoordinate(mesh)

    # exact solution and forcing interpolated into V
    u_exact = Function(V, name="u_exact").interpolate(exact_expression(x, y))
    f = Function(V, name="forcing").interpolate(forcing_expression(x, y))

    # Method 1: standard weak-form FEM

    # step 1:
    u = TrialFunction(V) # The unknown or variable u(x,y)
    v = TestFunction(V) # The testfunction of u

    #step 2:
    #here we are not using fem expansion
    a = inner(grad(u), grad(v)) * dx # first term of the weak form
    L = f * v * dx # second term of the weak form

    #step 3:
    #FEM expansion using u~u_1
    u_1 = Function(V, name="u_1")

    #Dirichlet boundary conditions at x=0,1 are imposed using,
    bc_x0 = DirichletBC(V, Constant(0.0), 1)
    bc_x1 = DirichletBC(V, Constant(0.0), 2)
    #No need to impose Neumann boundary conditions, as they are automatically included

    #step 4:
    #solving the algebraic system to find u_1
    solve(a == L, u_1, bcs=[bc_x0, bc_x1],
    | solver_parameters={"ksp_type": "cg", "pc_type": "hypre" }
    )

    # Method 2: Ritz-Galerkin / energy minimisation

    #step 1:
    #solution u_2 that minimises the functional J[u] = Ju
    u_2 = Function(V, name="u_2")

    #step 2:
    # using the same test function v
    Ju = (0.5 * inner(grad(u_2), grad(u_2)) - u_2 * f) * dx

    #step 3:
    #derivative of the functional with respect to u_2 in the direction of v
    F = derivative(Ju, u_2, du=v)

    #step 4:
    #solving F=0 to find u_2
    solve(F == 0, u_2, bcs=[bc_x0, bc_x1],
    | solver_parameters={"ksp_type": "cg", "pc_type": "hypre"}
    )

```

Figure 5: code snapshot

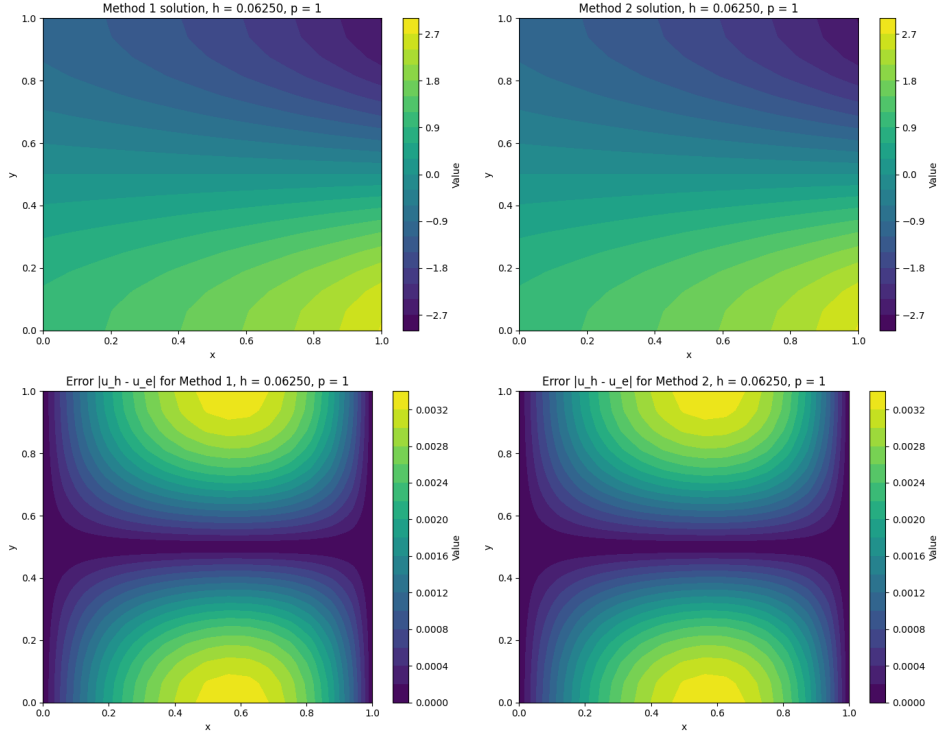


Figure 6: for  $p=1$  with modified BCs

Therefore, the forcing term becomes

$$f(x, y) = -\Delta u = (\pi^2 - 1)e^x \cos(\pi y)$$

The non-homogeneous Dirichlet boundary conditions for these are,

$$u(0, y) = \cos(\pi y), \quad u(1, y) = e \cos(\pi y)$$

But the horizontal boundaries satisfy homogeneous Neumann conditions,

$$\frac{\partial u}{\partial y} = 0 \quad \text{at } y = 0, 1$$

As Neumann conditions are automatically incorporated in the weak formulation, therefore we are not explicitly introducing them in our code.

The results for polymer degree 1, 2 and 3 are shown in the figure 6, 7 and 8 respectively. And figure 9 presents the plots for the the L2error (how far is the numerical solution from the exact solution) vs mesh resolutions for both the methods for  $p=1, 2$  and 3.

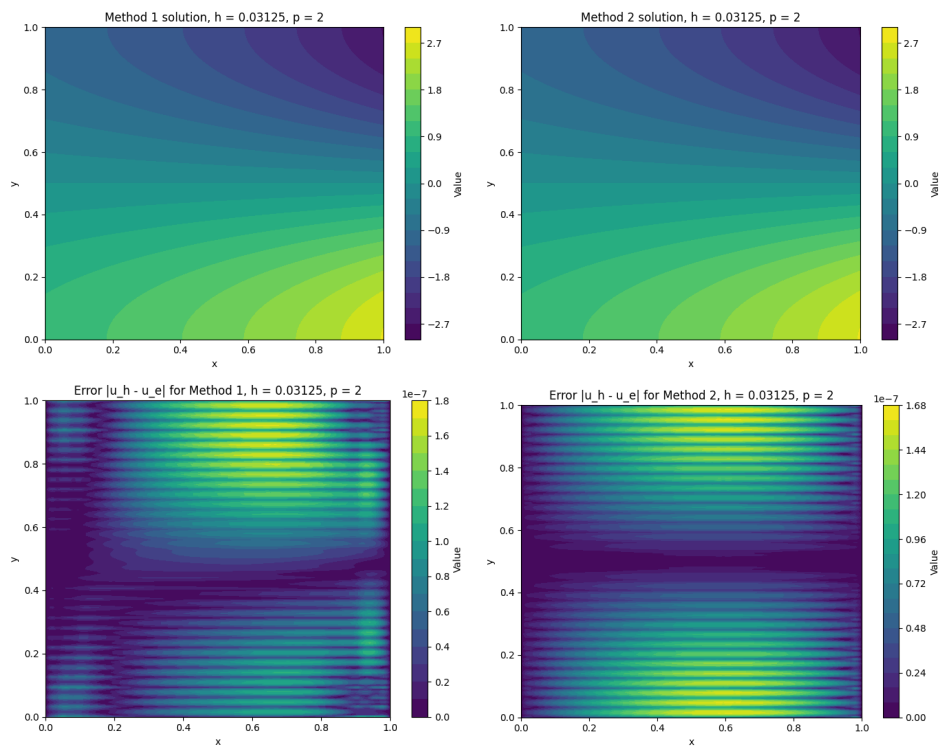


Figure 7: for  $p=2$  with modified BCs

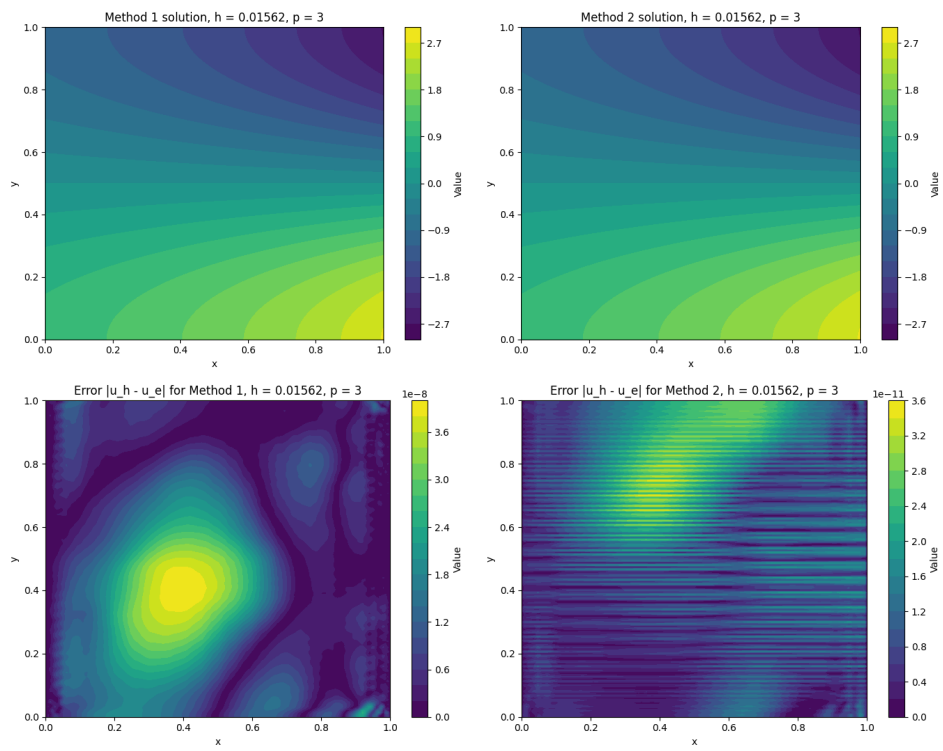


Figure 8: for  $p=3$  with modified BCs



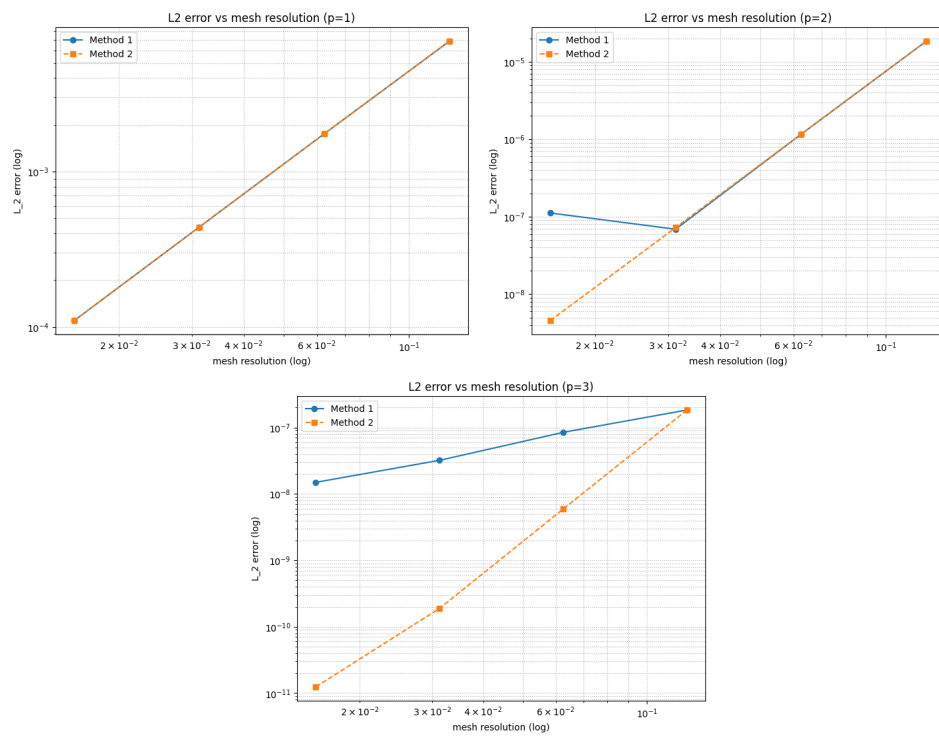


Figure 9: Error relation with the mesh resolutions for both the methods with the modified BC's.

## 2 Ground Water

We are considering one dimensional flow for groundwater, along the horizontal  $y$  direction. The groundwater height  $h_m(y,t)$  is the depth-averaged water storage, basically like a channel. Here, rainfall increases this groundwater height locally, and the water redistributes along the  $y$ -direction through a nonlinear diffusion mechanism. And at the boundary  $y=0$ , groundwater is coupled to a canal, where the water level adjusts dynamically.

The following equations describes the system:

$$\partial_t(w_v h_m) - \alpha g \partial_y (w_v h_m \partial_y h_m) = \frac{w_v R}{m_{por} \sigma_e}, \quad y \in [0, L_y]$$

The boundary conditions are,

$$\partial_y h_m = 0 \quad \text{at } y = L_y, \quad h_m(0, t) = h_{cm}(t)$$

and the canal equation is,

$$L_c w_v \frac{dh_{cm}}{dt} = w_v m_{por} \sigma_e \frac{1}{2} \alpha g \partial_y (h_m^2) \big|_{y=0} - w_v \sqrt{g} \max \left( \frac{2}{3} h_{cm}, 0 \right)^{3/2}$$

### 2.1 Explicit finite element and time discretisation of the groundwater canal model

#### Weak formulation

$w_v$  is constant and nonzero. So, we are dividing the groundwater equation by  $w_v$ . So the equation becomes:

$$\partial_t(h_m) - \alpha g \partial_y (h_m \partial_y h_m) = \frac{R}{m_{por} \sigma_e}$$

Now multiplying this by a test function  $q(y)$ , we integrate over the domain:

$$\int_0^{L_y} q \partial_t h_m dy - \alpha g \int_0^{L_y} q \partial_y (h_m \partial_y h_m) dy = \int_0^{L_y} q \frac{R}{m_{por} \sigma_e} dy$$

Using integration by parts to solve the nonlinear diffusion term

$$\int_0^{L_y} q \partial_t h_m dy + \alpha g \int_0^{L_y} h_m (\partial_y q) (\partial_y h_m) dy - \alpha g [q h_m (\partial_y h_m)]_0^{L_y} = \int_0^{L_y} q \frac{R}{m_{por} \sigma_e} dy$$

Now applying the boundary conditions,

No flux condition  $\partial_y h_m = 0$  at  $y = L_y$ , therefore, it vanishes. And the remaining boundary flux term at  $y = 0$  is used. Therefore, the equation becomes,

$$\int_0^{L_y} q \partial_t h_m dy + \alpha g \int_0^{L_y} h_m (\partial_y q) (\partial_y h_m) dy + \alpha g q(0) h_m(0, t) (\partial_y h_m(0, t)) = \int_0^{L_y} q \frac{R}{m_{por} \sigma_e} dy$$

As, we can write

$$h_m \partial_y h_m = \frac{1}{2} \partial_y (h_m^2)$$

Therefore, the canal equation can be rewritten as

$$L_c w_v \frac{dh_{cm}}{dt} = w_v m_{por} \sigma_e \frac{1}{2} \alpha g \partial_y (h_m^2) \big|_{y=0} - w_v \sqrt{g} \max \left( \frac{2}{3} h_{cm}, 0 \right)^{3/2}$$

and it becomes,

$$\frac{1}{2} \alpha g \partial_y (h_m^2) \big|_{y=0} = \frac{L_c}{m_{por} \sigma_e} \frac{dh_{cm}}{dt} + \frac{\sqrt{g}}{m_{por} \sigma_e} \max \left( \frac{2}{3} h_{cm}, 0 \right)^{3/2}$$

Now we are substituting this into the weak formulation and using the coupling condition  $h_m(0, t) = h_m(t)$ .

$$\begin{aligned} & \int_0^{L_y} q \partial_t h_m dy + q(0) \frac{L_c}{m_{por} \sigma_e} \frac{dh_{cm}}{dt} = \\ & \int_0^{L_y} \left( -\alpha g h_m (\partial_y q) (\partial_y h_m) + q \frac{R}{m_{por} \sigma_e} \right) dy - q(0) \frac{\sqrt{g}}{m_{por} \sigma_e} \max \left( \frac{2}{3} h_{cm}, 0 \right)^{3/2} \end{aligned}$$

## Finite element discretisation

We are discretising the spatial domain using continuous piecewise linear finite elements (CG1).

Assuming  $\phi_j(y)_{j=1}^{N_n}$  as the piecewise linear continuous basis functions.

So the approximate groundwater height is,

$$h_m(y, t) \approx h_h(y, t) = \sum_{j=1}^{N_n} h_j(t) \phi_j(y)$$

with  $h_1(t) = h_m(0, t) = h_m(t)$ .

Choosing  $q = \phi_i$  as the test function.

The time term,

$$\int_0^{L_y} q \partial_t h_m dy \approx \int_0^{L_y} \phi_i \partial_t \left( \sum_j h_j \phi_j \right) dy = \sum_j \left( \int_0^{L_y} \phi_i \phi_j dy \right) \frac{dh_j}{dt}$$

Considering,

$$M_{ij} = \int_0^{L_y} \phi_i \phi_j dy$$

Therefore time term becomes,  $M_{ij} \frac{dh_j}{dt}$   
the semi-discrete system becomes

$$M_{ij} \frac{dh_j}{dt} + \delta_{i1} \frac{L_c}{m_{por} \sigma_e} \frac{dh_1}{dt} = b_i - \delta_{i1} \frac{\sqrt{g}}{m_{por} \sigma_e} \max \left( \frac{2}{3} h_1, 0 \right)^{3/2}$$

where the nonlinear load vector is

$$b_i = \int_0^{L_y} \left( -\alpha g h_h (\partial_y \phi_i) (\partial_y h_h) + \phi_i \frac{R}{m_{por} \sigma_e} \right) dy$$

## Explicit time discretisation

We are using a forward Euler scheme for time discretisation. Let,  $h_j^n \approx h_j(t^n)$ .

Then

$$\frac{dh_j}{dt} \approx \frac{h_j^{n+1} - h_j^n}{\Delta t}$$

From the weak form, we get

$$M_{ij} \frac{h_j^{n+1} - h_j^n}{\Delta t} + \delta_{i1} \frac{L_c}{m_{por} \sigma_e} \frac{h_1^{n+1} - h_1^n}{\Delta t} = b_i^n - \delta_{i1} \frac{\sqrt{g}}{m_{por} \sigma_e} \max \left( \frac{2}{3} h_1^n, 0 \right)^{3/2}$$

After solving this, it becomes,

$$M_{ij} h_j^{n+1} + \delta_{i1} \frac{L_c}{m_{por} \sigma_e} h_1^{n+1} = M_{ij} h_j^n + \delta_{i1} \frac{L_c}{m_{por} \sigma_e} h_1^n + \Delta t b_i^n - \Delta t \delta_{i1} \frac{\sqrt{g}}{m_{por} \sigma_e} \max \left( \frac{2}{3} h_1^n, 0 \right)^{3/2}$$

For a linear diffusion system  $u_t = Du_{yy}$  the finite difference time stability is nearly,

$$\Delta t \leq \frac{(\Delta y)^2}{2D}$$

Since for our case the scheme is explicit and the governing equation is non-linear diffusion, so the stability requires a time step restriction analogous to the finite difference case.

Therefore the effective diffusivity scales as  $D_{\text{eff}} \sim \alpha g h_m$  and thus

$$\Delta t \leq \frac{(\Delta y)^2}{2\alpha g h_{\text{max}}}$$

where  $h_{\text{max}}$  is an the maximum groundwater height estimate.

## 2.2 Explicit finite element and time discretisation of the groundwater canal model

Simulations are performed using the parameter values

$$m_{\text{por}} = 0.3, \quad \sigma_e = 0.8, \quad L_y = 0.85, \text{ m}, \quad k = 10^{-8}, \text{ m}^2, \quad L_c = 0.05, \text{ m},$$

$$R_{\text{max}} = 1.25 \times 10^{-4}, \text{ m/s}, \quad \nu = 10^{-6}, \text{ m}^2/\text{s}, \quad g = 9.81, \text{ m/s}^2.$$

Also, simulations are performed on meshes with (m=10, 20, 40, 80) elements.

Figure 10 and 11 shows the results for the fixed depth of 0.07m using CG1. Figure 10 presents the results when mesh elements are 10 (left side plots) when mesh elements are 20 (right side plots). Whereas, figure 11 presents the results when mesh elements are 40 (left side plots) when mesh elements are 80 (right side plots). And the groundwater profiles increase monotonically in time, it shows the diffusive redistribution of water within the porous medium. As time progresses, the profiles gradually approach a quasi-steady configuration.

Figure 12 and 13 are for variable canal height using CG1. Figure 12 presents the results when mesh elements are 10 (left side plots) when mesh elements are 20 (right side plots). Whereas, figure 13 presents the results when mesh elements are 40 (left side plots) when mesh elements are 80 (right side plots). These results presents a different behaviour in comparision of previous. Starting from zero, it increases due to inflow from the groundwater at (y=0). A periodic rainfall pattern is used, it mean rain may or may not happen based upon these periods. Periods when rainfall happens, a rapid increases in groundwater levels and enhanced inflow to the canal will takes place. On the other hand, for dry intervals, reduce the inflow reduces and outflow dominate. So basically there is a competition between inflow and nonlinear outflow to approaching a statistical steady state.

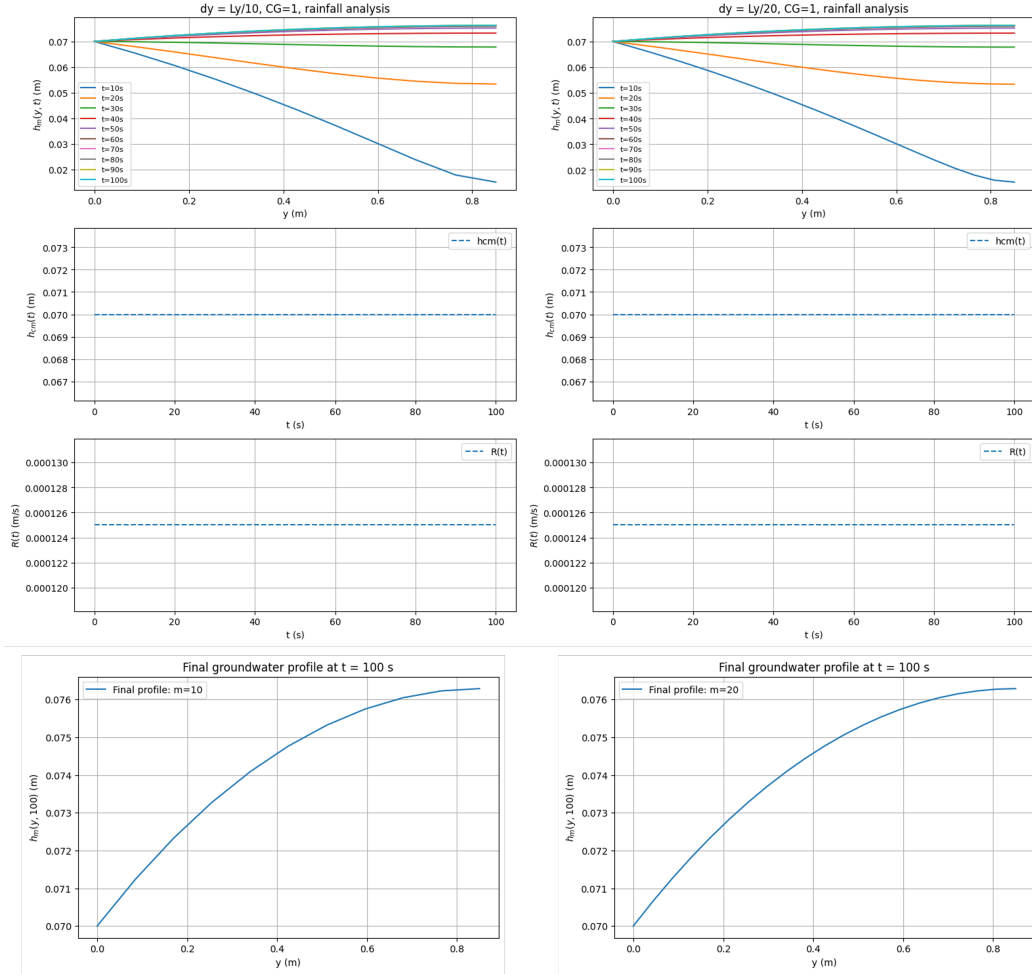


Figure 10: CG1 for fixed depth of 0.07m. Left hand side plots are for  $m=10$  and right hand side plots are for  $m=20$ .

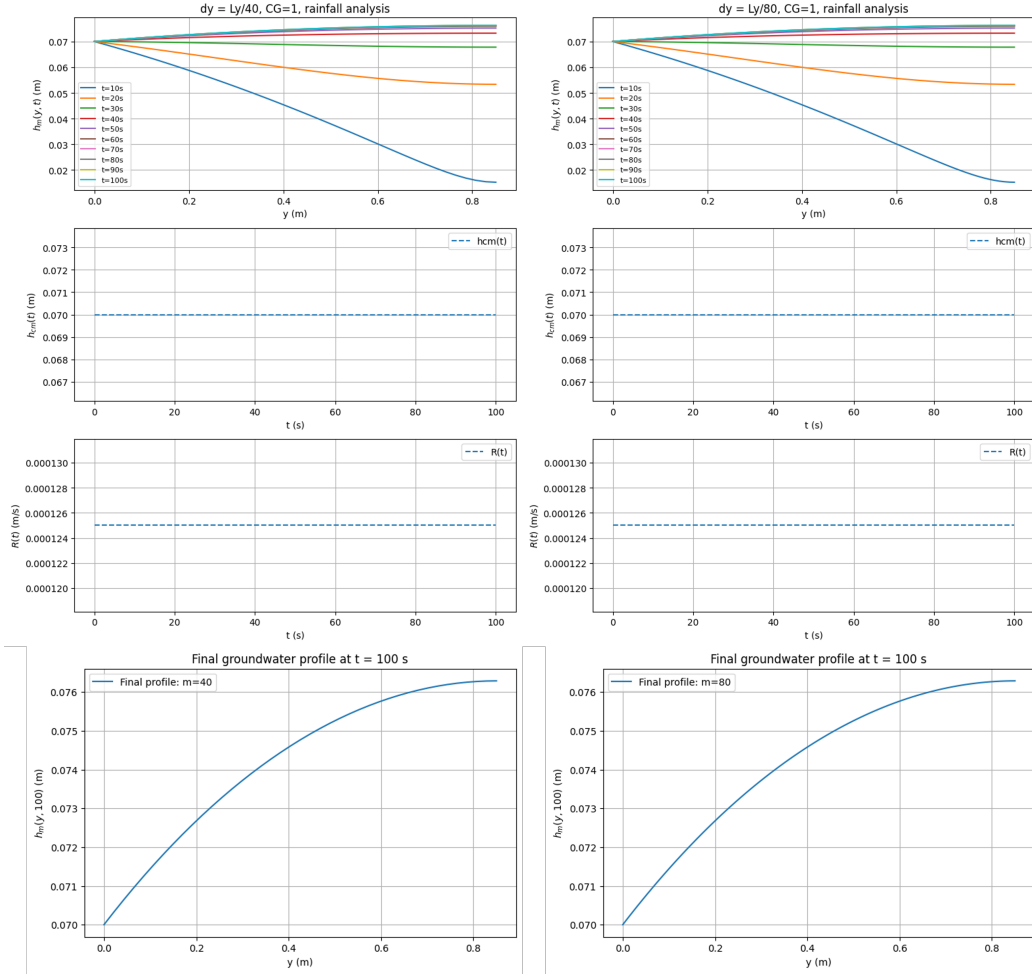


Figure 11: CG1 for fixed depth of 0.07m. Left hand side plots are for  $m=40$  and right hand side plots are for  $m=80$ .

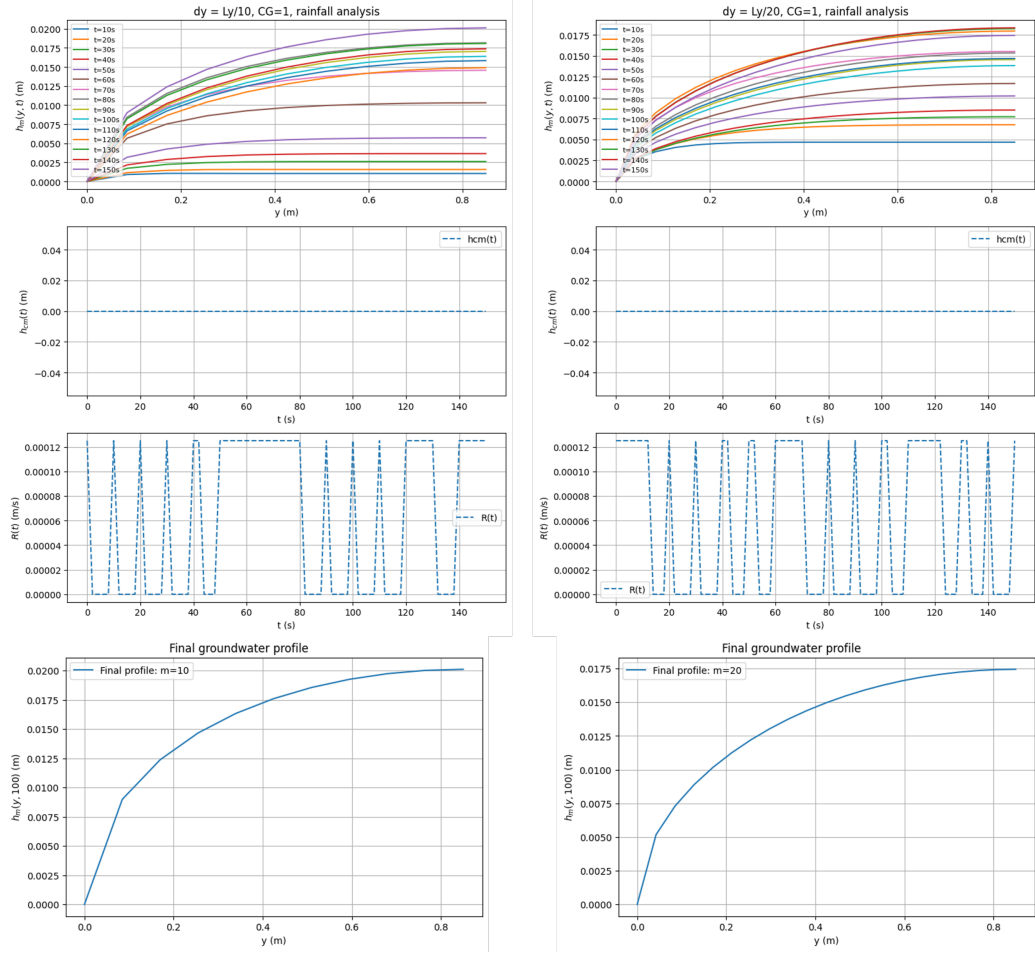


Figure 12: CG1 for changing depth. Left hand side plots are for  $m=10$  and right hand side plots are for  $m=20$ .



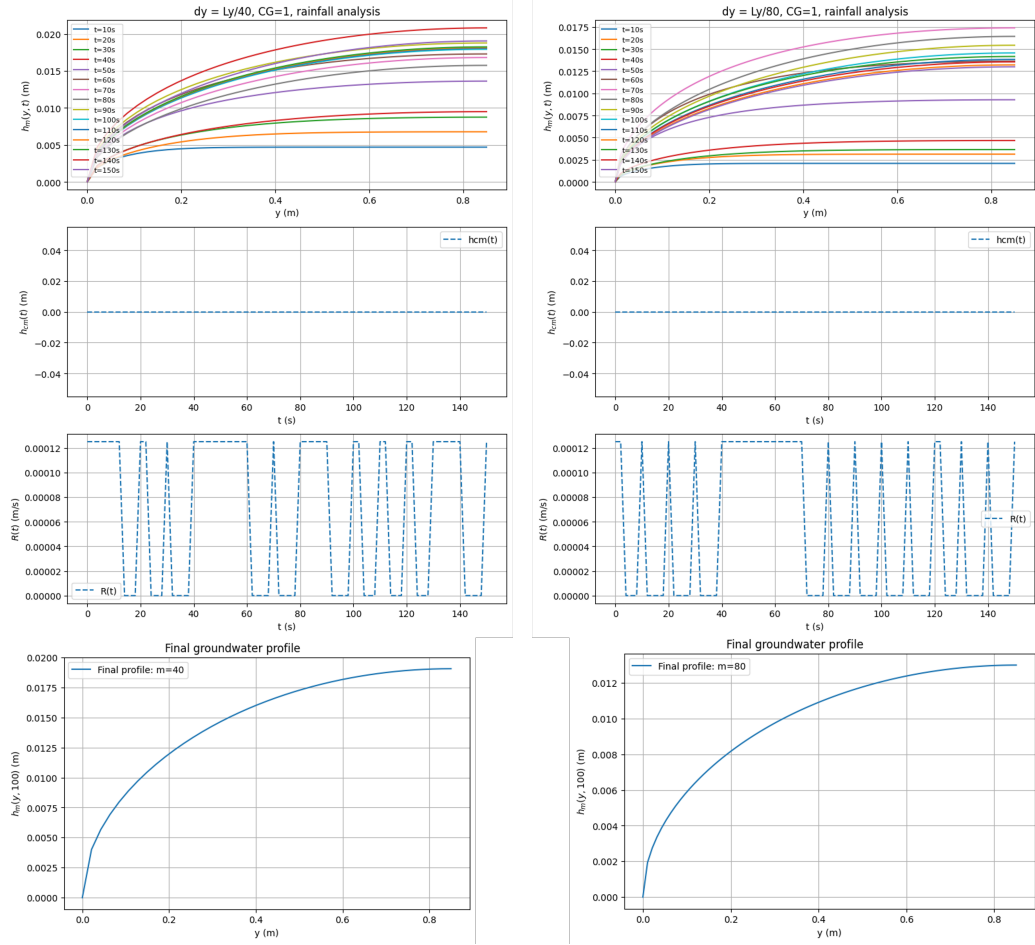


Figure 13: CG1 for changing depth. Left hand side plots are for  $m=40$  and right hand side plots are for  $m=80$

## 2.3 Crank-Nicolson scheme

The coupled groundwater canal system given by

$$\partial_t h_m - \alpha g \partial_y (h_m \partial_y h_m) = \frac{R(t)}{m_{por} \sigma_e}, \quad y \in (0, L_y), \quad t > 0$$

with boundary condition

$$h_m(0, t) = h_{cm}(t)$$

and a natural (no-flux) condition is used at  $y = L_y$ .

The equation for canal height change is,

$$\frac{L_c}{m_{por} \sigma_e} \frac{dh_{cm}}{dt} = h_m(0, t) - \sqrt{g} \max\left(\frac{2}{3} h_{cm}, 0\right)^{3/2}$$

The weak form in the finite element space  $V$  for CG is,

$$\int_0^{L_y} \partial_t h_m \phi \, dy + \int_0^{L_y} \alpha g h_m \partial_y h_m \partial_y \phi \, dy = \int_0^{L_y} \frac{R(t)}{m_{por} \sigma_e} \phi \, dy$$

## Crank–Nicolson time discretisation

We use Crank–Nicolson scheme to obtain a second-order accurate time discretisation. Let  $t^n = n\Delta t$ . And the time derivative is approximated by,

$$\partial_t h_m \approx \frac{h^{n+1} - h^n}{\Delta t}$$

evaluating the nonlinear diffusion term at the midpoint in time.

$$h_m \partial_y h_m \approx \frac{1}{2} (h^{n+1} \partial_y h^{n+1} + h^n \partial_y h^n)$$

The fully discrete weak form,

$$\int_0^{L_y} \frac{h^{n+1} - h^n}{\Delta t} \phi \, dy + \frac{1}{2} \int_0^{L_y} \alpha g (h^{n+1} \partial_y h^{n+1} + h^n \partial_y h^n) \partial_y \phi \, dy = \int_0^{L_y} \frac{R^{n+1/2}}{m_{por} \sigma_e} \phi \, dy$$

The term

$$h^{n+1} \partial_y h^{n+1}$$

introduces nonlinearity into the system. Therefore, at every time step we need solve a nonlinear algebraic system of the form

$$F(h^{n+1}) = 0$$

We are using Newton's method to solve the nonlinear system. The initial guess for that is  $h^{(0)}$ , and the iteration proceeds as

$$J(h^{(k)}) \delta h^{(k)} = -F(h^{(k)})$$

$$h^{(k+1)} = h^{(k)} + \delta h^{(k)}$$

where  $J(h)$  is the Jacobian of  $F$ .

And the iteration will continue until

$$\|F(h^{(k)})\| < \varepsilon$$

where,  $\varepsilon$  is the tolerance.

The discretised canal equation is,

$$h_{cm}^{n+1} = h_{cm}^n + \frac{\Delta t}{\gamma} \left[ h_m^n(0) - \sqrt{g} \max\left(\frac{2}{3}h_{cm}^n, 0\right)^{3/2} \right]$$

where

$$\gamma = \frac{L_c}{m_{por}\sigma_e}$$

Figure 14 presents the results when Crank–Nicolson scheme is used with CG=1.

## 2.4 CG3 with and without Stabilization

Figure 15 shows the results when Crank–Nicolson scheme is used with CG=3. And figure 16 presents the results when stabilization term is used, as shown in the code.

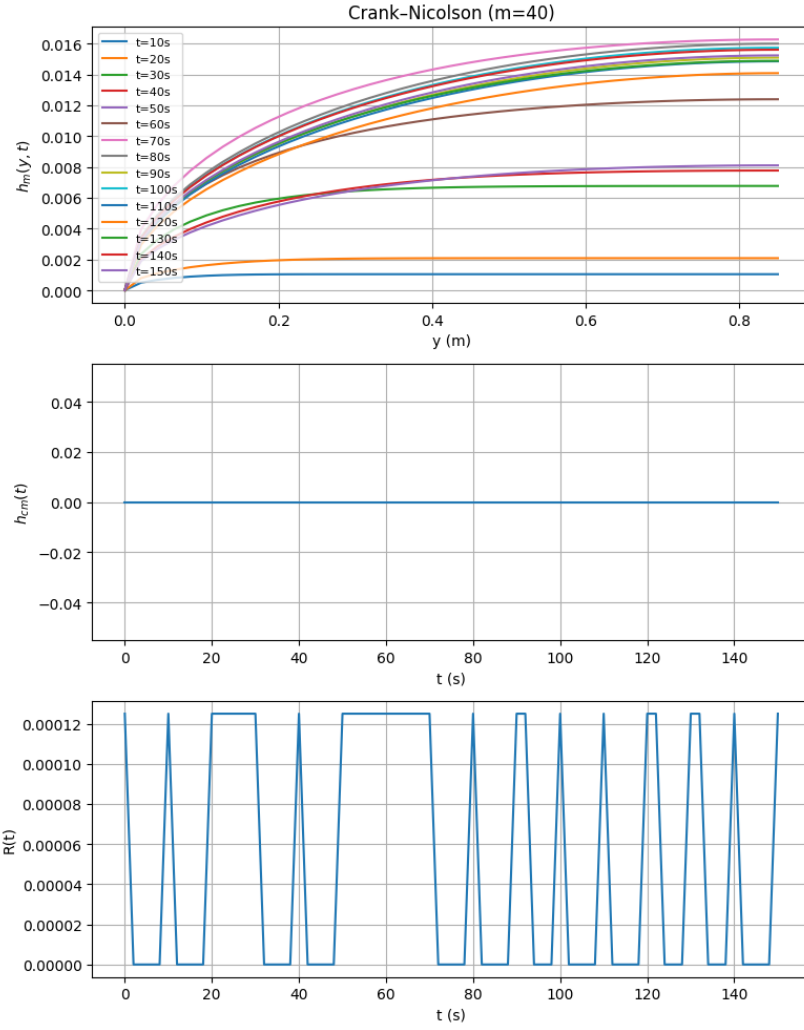


Figure 14: plots for the Crank–Nicolson scheme with CG=1

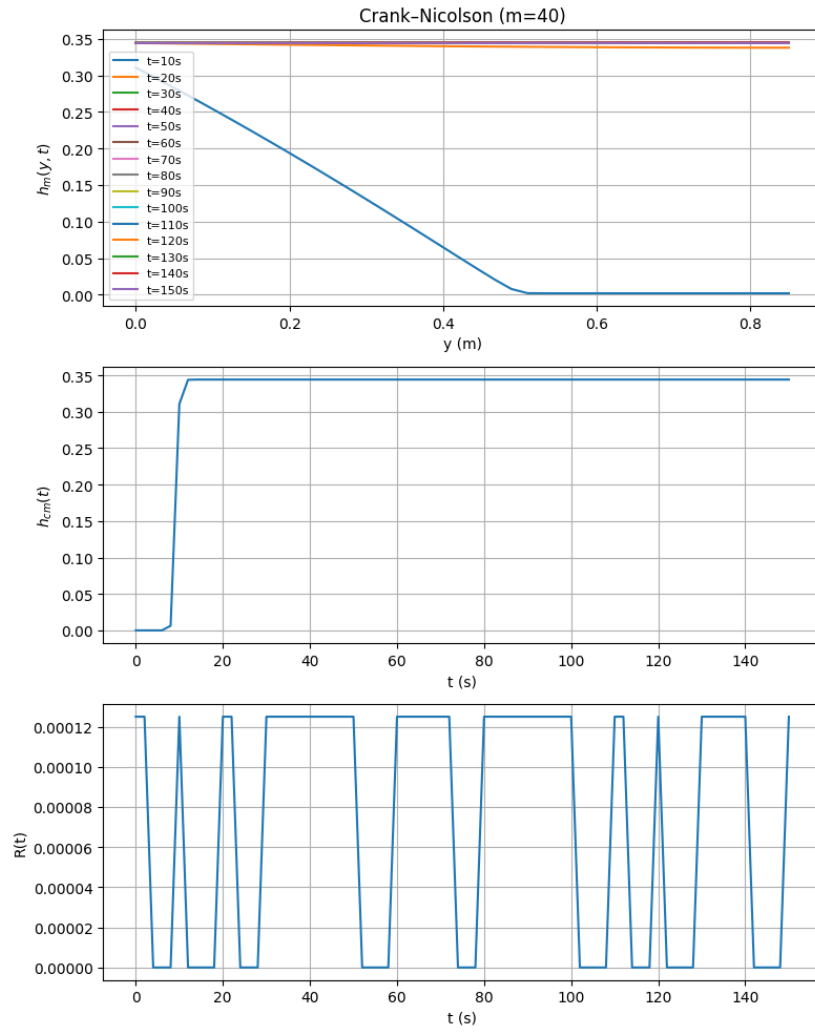


Figure 15: plots for the Crank–Nicolson scheme with CG=3.

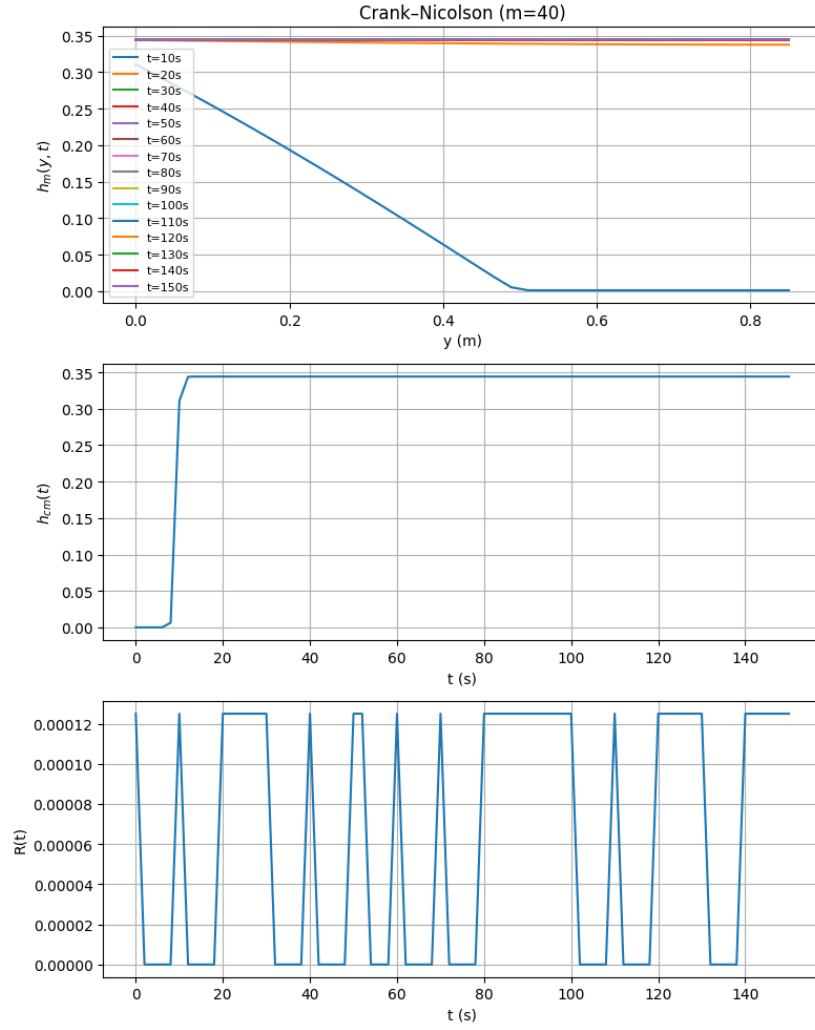


Figure 16: plots for the Crank–Nicolson scheme with CG=3 with the stabilization term.